Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 183 (2021) 281–287

Procedia Computer Science

www.elsevier.com/locate/procedia

10th International Conference of Information and Communication Technology (ICICT-2020)

A novel reasoning model for credit investigation system based on Fuzzy Bayesian Network

Hui-Dong Wu, Lu Han*

School of Management Science and Engineering, Central University of Finance and Economics, Beijing 100080, China

Abstract

Nowadays, credit is becoming more and more important and the progress of artificial intelligence technology brings new opportunities for the development of credit investigation. In this paper, we come up with a novel reasoning model for a credit investigation system based on the fuzzy Bayesian network which is a new convergence of technology and finance. In the model, combined with data learning and expert knowledge systems, a new knowledge representation method is proposed according to fuzzy theory and transfer rule of node value is defined. The problem that continuous value cannot be used in the traditional Bayesian network is well solved and the reasoning process is more suitable for credit characteristics. The most important innovations of our model are a new conditional probability function and two data fusion algorithms, which make fuzzy numbers well transmit and merge. What's more, the approach makes full use of the advantages of big data and its rationality and stability are verified. In practice, the proposed model has significant meaning for the development of investigation systems and it can provide reference value for the intelligence of other financial fields.

Keywords: fuzzy Bayesian network; explainable artificial intelligence; credit investigation; causal inference

1. Introduction

With the development of economic globalization and technology, there is a great expansion of information in the business field[1]. On the one hand, the social networks are becoming more and more complex thanks to the Internet, causing a severer result of information asymmetry. On the other hand, a variety of financial products are emerged,

* *Corresponding E-mail address:* hanluivy@126.com

10.1016/j.procs.2021.02.060

which adds to the difficulties of market regulation. Therefore, it is of great significance to accelerate the construction and improvement of credit investigation systems.

In the last few years, artificial intelligence has provided a new research direction, and so far, the credit investigation has become a typical example of the close combination of information technology and finance[2]. However, breakthroughs in efficiency and accuracy alone are not the whole story of AI. As a powerful instrument, causal inference is applied to traditional algorithms, forming a new field called explainable artificial intelligence (XAI) which is an important step towards the era of AI 2.0[3].

In this paper, in order to improve the efficiency of reasoning process in credit investigation systems, we proposed a novel reasoning model using fuzzy Bayesian network based on XAI theory. Firstly, a new knowledge representation method is proposed. Then the conditional probability function and data fusion algorithms are defined so that the fuzzy parameter with new form can transmit and merge well. The running efficiency of the model is very high and the results are very good, which means our work is novel and effective.

In the second section, the existed researches will be reviewed; our model will be built in section 3; section 4 is for case study; and the last part is the conclusion of this paper.

2. Literature Review

Bayesian network (BN) was first proposed by Pearl[4] in 1980s. Because of its powerful functions on causal inference and the probability field, the instrument is widely used in reasoning decision researches[5]. In recent years, in view of the modelling role of Bayesian networks in the field of uncertainty, this method, combined with the knowledge of probability theory and graph theory, has been widely promoted in artificial intelligence. So far, the Bayesian network has proved its value in many aspects[6], including medical diagnosis, machine fault diagnosis, user modelling, fraud identification and so on.

However, in the aspect of uncertainty research, Bayesian network can only solve the reasoning problem by means of probability calculation. In general, the fuzzy expression in daily life cannot be used as the data source and basis of Bayesian network, and the continuous variables cannot be accurately reflected. However, the fuzzy theory just makes up for the deficiency in this aspect. Fuzzy Logic is a method proposed by Zadeh[7], which transforms general Fuzzy expressions through membership functions, making it possible to objectively reflect language uncertainty.

The essence of fuzzy Bayesian network is to combine fuzzy logic theory and traditional Bayesian network to solve the problem on continuous values. In existing researches, the fuzzy Bayesian network has been successfully applied in the fields of data mining, confidence propagation and industrial engineering. Tang and Liu[8] solved a machine fault diagnosis problem with fuzzy events and the Bayesian network, while there are only binary values in their model. Zhou[9] set a fuzzy Bayesian model called CREAM to reduce the bad influence of subjective cognition. Ren[10] analyse an offshore risk using fuzzy Bayesian network. In order to analyse the dynamic safety of chemical process systems, Akbar[11] proposed a novel fuzzy Bayesian network.

As we can see, the fuzzy Bayesian network is mainly applied to control risks for certain field, but it is rarely used in finance. Our work is to introduce this method to credit investigation systems and make our proposed model available.

3. Reasoning Model for Credit Investigation System

In this section, we will construct the reasoning model for credit investigation systems. The model is composed of two parts, the Bayesian inference framework and the setting of specific parameters. In order to fuzzy the parameters, some theories are referred, including fuzzy C-means[12], membership matrix[13] and triangular fuzzy number[14]. Then a new transfer rule of fuzzy numbers in Bayesian networks is proposed. Fig.1 is the procedure of our model.

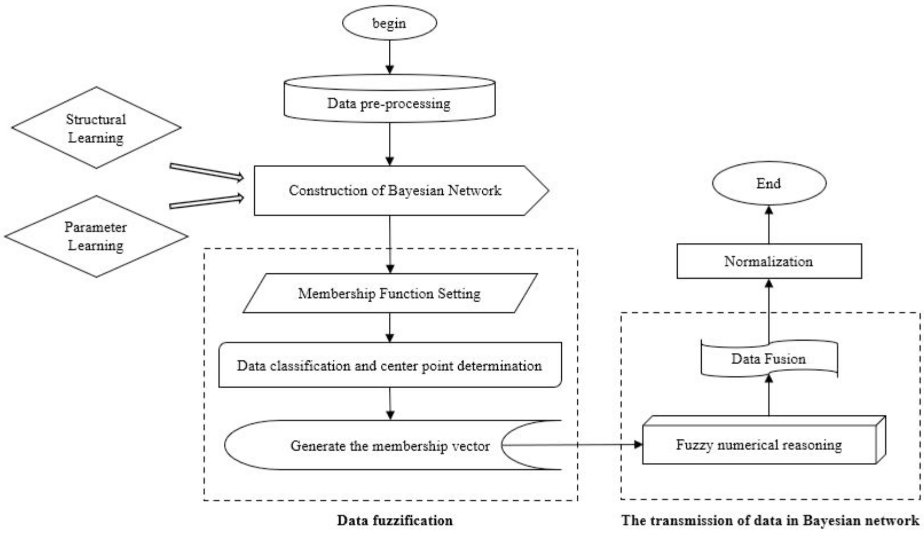

Fig. 1. Procedure of the reasoning model for credit investigation system

3.1. Construction of Bayesian Network

The construction of a Bayesian network in a certain field usually consists of three steps: the identification of influencing variables and their possible values, the identification and visualization of inter-variable dependencies, and the learning of inter-variable parameters[15]. In addition to possible values identification in the first two, others can be classified as structural learning[16], and the rest belongs to parameter learning[17].

In order to better represent the relationship between nodes and make the information in the Bayesian network fully reflect the real situation, we also introduce a tag representing the positive and negative influence of the relationship in this model, that is, the judgment of whether the parent node has a positive or negative influence on the child node. The tag avoids the neglect of the mutual inhibition between nodes and makes the inference result of the model more fit to the reality. A simple Bayesian network that conforms to our model is shown in the Fig.2.

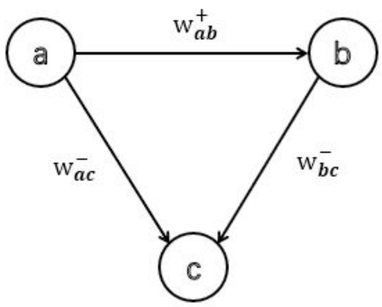

Fig. 2. A simple Bayesian Network

3.2. Parameter fuzzification

As for fuzzification of node value, we first select the normal function as the membership function according to the characteristics of credit system. Then FCM is utilized as the approach of data learning to calculate central nodes for different classes in membership function. Moreover, the strengths of relationships between nodes are expressed as some triangular fuzzy numbers on the basic of the results of parameter learning. The specific steps are as follows.

For each node, set the membership function as shown in the formula (1).

$$\theta_i(x,\sigma_i) = e^{-\frac{(x-v_i)^2}{2\sigma_i^2}}, i = 1,2,3 \tag{1}$$

In the formula (1), θ_i represents the fuzzy value for class i. The letter x is the original node value. The symbol σ means the degree of uncertainty and v_i is the central point of class i. To match the triangular fuzzy numbers for subsequent reasoning, the quantity of classes in this model is set as 3.

Then the unknown parameters in formula (1) are determined by the way of data learning. For the calculation of the center point v of different grades, the fuzzy C-means is adopted. The goal of FCM is to divide the data set containing n subjects $X=\{x_1,\dots x_n\}$ into C clusters by minimizing the objective function (2), and the central point can be expressed as $V=\{v_1,\dots v_C\}$.

$$J_m = \sum_{i=1}^{n}\sum_{j=1}^{C}(u_{ij})^m \| x_i - v_j \|^2 \tag{2}$$

In function (2), m is the number of clusters and it is assigned to 3. The letters i and j are labels of different clusters. The v_j is the central point of j-th cluster and $u_j \in [0,1]$ can be described as the degree of membership, $\|\cdot\|$ is for distance calculation.

The optimal solution can be obtained after several iterations. We call the three clusters low risk set, medium risk set and high risk set respectively in line with their characteristics, and calculate their degree of uncertainty σ with the formula (3) where m_i is the quantity of data of cluster i and N_i is the total quantity of data sets.

$$\sigma_i = \frac{m_i}{N_i}(i = 1,2,3) \tag{3}$$

What needs to illustrate is that there may be binary values. On this occasion, 0 is represented as (0.9,0.1,0.1) while (0.1,0.1,0.9) is the membership vector of number 1.

To sum up, the value of i-th node is transformed into a membership vector $\overline{\theta}_i = (\theta_{i1}, \theta_{i2}, \theta_{i3})$.

For the relationships between every two nodes, we use w that is calculated in parameter learning step and transformed it into a triangular fuzzy number. There are five classes being set which are low correlation, medium low correlation, medium correlation, medium high correlation and high correlation. In this step, tags will not change. Table 1 shows specific fuzzy numbers with positive tags as example.

Table 1. Triangular fuzzy numbers of different correlation strengths

Correlation Strength	Triangular Fuzzy Number
Low Correlation	$(0.1,0.1,0.25)^+$
Medium Low Correlation	$(0.1,0.25,0.5)^+$
Medium Correlation	$(0.25,0.5,0.75)^+$
Medium High Correlation	$(0.5,0.75,0.9)^+$
High Correlation	$(0.75,0.9,0.9)^+$

3.3. Value Transfer

Now the value of parent nodes and correlation between nodes are all transformed into fuzzy numbers. In order to realize the causal inference in the fuzzy Bayesian network, it is necessary to define passing rules of fuzzy value. In our model, we define a conditional probability function to get the reasoning results of child nodes as the formula (4).

$$\theta(y_k \mid i=\overline{\theta}_i)=\begin{cases}\dfrac{\omega_{j1}\theta_{i1}-y_k}{\theta_{i1}\omega_{j2}+\omega_{j1}\theta_{i2}},\omega_{j1}\theta_{i1}-(\theta_{i1}\omega_{j2}+\omega_{j1}\theta_{i2})\le y_k<\omega_{j1}\theta_{i1}\\ \dfrac{y_k-\omega_{j1}\theta_{i1}}{\theta_{i1}\omega_{j3}+\omega_{j1}\theta_{i3}},\omega_{j1}\theta_{i1}\le y_k\le\omega_{j1}\theta_{i1}+(\theta_{i1}\omega_{j3}+\omega_{j1}\theta_{i3})\\ 0.5,other\end{cases} \tag{4}$$

Thereinto, $\boldsymbol{\theta_y}=(\theta_{y1},\theta_{y2},\theta_{y3})$ is the reasoning result of the i-th child node and k is the label of factors in the vector. $\boldsymbol{\omega_y}=(\omega_{y1},\omega_{y2},\omega_{y3})$ represents the triangular fuzzy number. What's more, we set y_k as 0.1,0.5 and 0.9 in order to calculate different θ_{yk}, and the value of y_k is always between 0 and 1.

Further on, if the tag of $\boldsymbol{\omega_y}$ is positive, the reasoning results will not change. Otherwise, $\theta_y=1-\theta_y$.

3.4. Data Fusion

Generally, there are more than two parent nodes for every child node, and the reasoning results obtained from different parent nodes are likely to be different[18]. Therefore, data fusion of different causal inference results into a single membership vector is a problem that must be solved in the process of data transmission in the fuzzy Bayesian network.

Considering that there may be conflicts between the child node values inferred by different parent nodes, the fuzzy data is first modified before data fusion. Algorithm1 is the process of conflict identification and data modification.

Algorithm1: Conflict Identification and Data Modification
Input: Matrix $\boldsymbol{G_{3\times n}}$ composed of n membership vectors inferred by different parent nodes
Output: Matrix $\boldsymbol{\breve{G}}$ composed of modified data
• Normalize the matrix $\boldsymbol{G}$ by columns to get the normalized matrix $\boldsymbol{G'}$; • Calculate the average number of each row $\boldsymbol{q_i}=\frac{1}{n}\sum_{j=1}^{n}\boldsymbol{G'_{ij}}$ （i=1,2,3） for $\boldsymbol{G'}$ and construct the average vector $\boldsymbol{Q}=(\boldsymbol{q_1},\boldsymbol{q_2},\boldsymbol{q_3})$; • Calculate the difference value between each membership vector and the average vector $\boldsymbol{d_j}=\sum\lvert\boldsymbol{G'_{\cdot j}}-\boldsymbol{Q}\rvert,(\boldsymbol{j}=\boldsymbol{1},\boldsymbol{2},\ldots,\boldsymbol{n})$in turn; • Calculate assigned weights $\boldsymbol{r_j}=\frac{\boldsymbol{p}}{\boldsymbol{sum}(\mathbf{p})}$ for each column where $p=1/d_i$; • If $\boldsymbol{r_j}<\frac{\boldsymbol{1}}{\boldsymbol{n}}$, membership vector of the j-th column is updated with the formula $\boldsymbol{G'_{\cdot j}}=\frac{r_j}{max(r_j)}\cdot\boldsymbol{G'_{\cdot j}}$,and $\boldsymbol{G'_{4j}}=\boldsymbol{1}-\frac{r_j}{max(r_j)}$,the rest of the row is set as 0; otherwise, do not modify.

In algorithm1, each value in the newly added row represents the uncertainty of the corresponding column.

The data fusion method using modified data is shown in algorithm 2.

Algorithm2: Fusion Process Based on Modified Data

Input: Matrix $\breve{\boldsymbol{G}}$ composed of modified data

Output: membership vector $\widetilde{\boldsymbol{\theta_y}}$ of child node

- Set the initial value of the variable t to 1;
- Calculate the normalization constant $\boldsymbol{K} = \sum_{i=1}^{3} \widetilde{\boldsymbol{G_{it}}} \cdot \widetilde{\boldsymbol{G_{i(t+1)}}} + \sum_{i=1}^{3} (\boldsymbol{G_{it}} \cdot \boldsymbol{G_{4(t+1)}} + \boldsymbol{G_{4t}} \cdot \boldsymbol{G_{i(t+1)}})$;
- Integrate t-th and t+1-th column of $\breve{\boldsymbol{G}}$, with the formula $\widetilde{\boldsymbol{G_{i(t+1)}}} = \widetilde{\boldsymbol{G_{it}}} \oplus \widetilde{\boldsymbol{G_{i(t+1)}}} = \frac{\mathbf{1}}{\boldsymbol{K}} (\widetilde{\boldsymbol{G_{it}}} \cdot \widetilde{\boldsymbol{G_{i(t+1)}}} + \widetilde{\boldsymbol{G_{it}}} \cdot \widetilde{\boldsymbol{G_{4(t+1)}}} + \widetilde{\boldsymbol{G_{4t}}} \cdot \widetilde{\boldsymbol{G_{i(t+1)}}})$;
- Set $t=t+1$ and repeat step 2-3 until $t=n$;
- Take the updated $\boldsymbol{G_{\cdot n}}$ which is the membership vector $\widetilde{\boldsymbol{\theta_y}}$.

In algorithm 2, the fusion process based on modified data is a pairwise merging process, which is fused into the membership vector of the merged child nodes through the gradual iteration and update of the values in the matrix. After normalizing the output of algorithm, the values can represent probabilities of different classes.

4. A Case Study

In order to verify the effectiveness of the proposed method, we select a commercial bank in China as the research object. In this part, according to the database and rule base of the commercial bank, the process of qualification verification of loan owners, one of the main processes of its lending business, is modeled.

Firstly, through the analysis of the data attributes and the content of the rule base in the database, the Bayesian causality model describing the determinants, causal factors, causal relationship and causal relationship intensity of the loan rejection result is extracted as Fig. 3 shows.

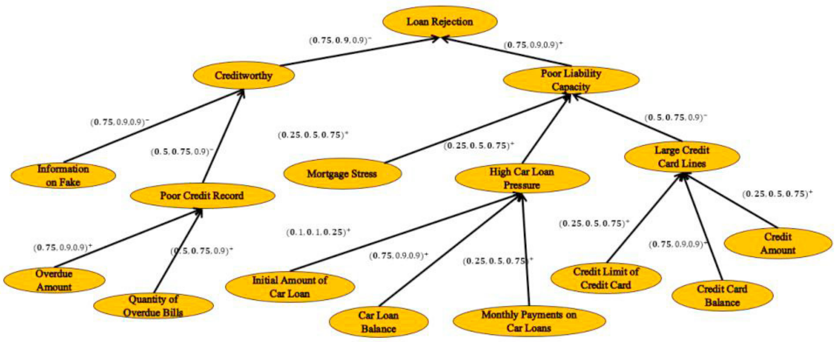

Fig. 3. Extracted Bayesian network from the process of qualification verification of loan owners

We select a sub-transfer process in Fig. 3 as an example for model application and calculation, and the numerical transfer between other nodes was consistent with this process, so it is unnecessary to elaborate.

Before the causal reasoning, the data to be judged should be fuzzy. We select a sub-DAG that contains overdue amount, quantity of overdue bills and poor credit record as an example. Combined with the existing database and expert system, we get (0.2, 0.4, 0.8) as overdue amount fuzzy numbers and (0.3, 0.7, 0.4) as quantity of overdue bills. Meanwhile, the intensities between nodes are obtained from data learning as $(0.75,0.9,0.9)^{+}$ and $(0.5,0.75,0.9)^{+}$. On this basis, causal reasoning and data fusion are carried out.

According to conditional probability function, the value of poor credit record is inferred from two paths respectively. When the parent node is overdue amount, the result of reasoning value is (0.1042,0.4487,0.9615). The other result is (0.1,0.9211,0.5). Because of the positive tags, the final vectors do not change.

Then it is time to integrate different values. The two membership vectors are substituted into algorithm 1 and algorithm 2, and the final membership vector is obtained as (0.0116,0.4547,0.5337). Therefore, if the manager of commercial bank set 0.5 as the threshold of loan rejection, the sample is defined as a poor credit record.

In order to further discuss the validity of the model, we also set up boundary points for calculation. The results showed good stability and were in line with general expectations.

5. Conclusion

In this paper, we proposed a fuzzy Bayesian network reasoning model through the analysis of the characteristics of the credit investigation system. Our model not only reduces the influence of subjective factors of the interference and solves the problem existed in traditional Bayesian network, but also makes the information contained in the network become richer. In a word, in practice, the model can meet the actual needs of application subjects and reflect the evaluation results more truly and objectively, which is of great practical significance for the further development of credit investigation system.

In the future researches, we will focus on the universality of this model[19-20]. For example, any quantity of level can be set according to the actual situation, rather than just three clusters. In that case, the conditional probability function should be further modified.

References

1. Wang YP. Analysis of Financial Business Model towards Big Data and its Applications. *J VCIR* 2019;**102729**:1047-3203.
2. Li F. *Innovation cases study of financial credit information driven by big data*; 2017.
3. Kuang K, Li L, Geng Z, Xu L. Causal Inference. *Engineering* 2019; **S2095-8099(19)**:30523-5
4. Pearl J. *Probabilistic reasoning in Intelligent systems: networks of plausible inference*; 1988.
5. Riggelsen C. Learning parameters of Bayesian networks from incomplete data via importance sampling. *Int J Approx Reason* 2006; **42**:69-83.
6. Wong ML, Guo YY. Learning Bayesian networks from incomplete databases using a novel evolutionary algorithm. *Support Syst* 2008;**45(2)** :368–383.
7. Zadeh LA. Fuzzy Sets. *Information and Control* 1965; **8(3)**:338-353.
8. Tang H, Liu S. Basic Theory of Fuzzy Bayesian Networks and Its Application in Machinery Fault Diagnosis. *FSKD* 2007.
9. Zhou Q, Wong YD, Loh HS, Yuen KF. A fuzzy and Bayesian network CREAM model for human reliability analysis– The case of tanker shipping. *Safety Science* 2018;**105**:149-157.
10. Ren J, Jenkinson I, Wang J. An Offshore Risk Analysis Method Using Fuzzy Bayesian Network. *J offshore mech arcti* 2009; **131(4)**:10-21.
11. Rostamabadi A, Jahangiri M, Zarei E, Kamalinia M, Alimohammadlou M. A novel Fuzzy Bayesian Network approach for safety analysis of process systems; An application ofHFACS and SHIPP methodology, *J Clean Pro* 2019; **S0959-6526(19)**:33631-5
12. Bezdek JC. Pattern Recognition with Fuzzy Objective Function Algorithms. *Kluwer Academic Publishers*, 1981.
13. Zhang QH, Wang J, Wang GY. The approximate representation of rough-fuzzy sets. *Chinese J Computers* 2015;**38(7)**:1484-1496.
14. Laarhoven PJMV, Pedrycz W. A Fuzzy Extension of Saaty's Priority Theory. 1983; **11(1-3)**:199-227.
15. Huang YP. Research on Bayesian Network. *Institute of Computing Technology, Chinese Academy of Sciences* 2005.
16. Li Z, Liu T, Zhang L. Structural Learningof Bayesian Network By Multi-population Bacterial Foraging Optimization. *IEEE ITOEC* 2018.
17. Zhou Y, Hospedales TM, Fenton N. When and where to transfer for Bayesian network parameter learning. *Expert Systems with Application* 2016;**55(aug.)**:361-373.
18. Andrade RL, Rêgo, Leandro C. The use of nodes attributes in social network analysis with an application to an international trade network. *Physica A* 2017;S0378437117308579.
19. Banihabib ME, Tanhapour M, Roozbahani A. Bayesian networks model for identification of the effective variables in the forecasting of debris flows occurrence. *Environmental Earth ences* 2020; **79(8)**.
20. Zhang H, Feng H, Cui Y. A Fuzzy Bayesian Network Model for Quality Control in O2O e-Commerce. *Int J C*CC 2020;**15(1)**.